14E022 Macroeconomics I

Practice Exam 2 — Answer Key

BSE — Academic Year 2025–26

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1 Question 1 — Augmented Solow Model with Human Capital

1. (5 points)

Solution:

Output per effective worker.

$$\tilde{y} = \frac{Y}{AL} = \frac{K^\alpha H^\beta (AL)^{1-\alpha-\beta}}{AL} = \frac{K^\alpha}{(AL)^\alpha} \cdot \frac{H^\beta}{(AL)^\beta} = \tilde{k}^\alpha \tilde{h}^\beta. \quad \checkmark$$

Law of motion for $\tilde{k}$.
Since $\tilde{k} = K/(AL)$, differentiate by the quotient/product rule:

$$\dot{\tilde{k}} = \frac{\dot{K}}{AL} - K\,\frac{\dot{A}L + A\dot{L}}{(AL)^2} = \frac{\dot{K}}{AL} - \tilde{k}\left(\frac{\dot{A}}{A} + \frac{\dot{L}}{L}\right) = \frac{\dot{K}}{AL} - (g+n)\,\tilde{k}.$$

Substituting $\dot{K} = s_K Y - \delta K$:

$$\dot{\tilde{k}} = \frac{s_K Y - \delta K}{AL} - (n+g)\,\tilde{k} = s_K \tilde{y} - \delta \tilde{k} - (n+g)\tilde{k}.$$

$$\boxed{\dot{\tilde{k}} = s_K\,\tilde{k}^\alpha \tilde{h}^\beta - (n+g+\delta)\,\tilde{k}.}$$

Law of motion for $\tilde{h}$. An identical calculation with $\dot{H} = s_H Y - \delta H$ gives:

$$\boxed{\dot{\tilde{h}} = s_H\,\tilde{k}^\alpha \tilde{h}^\beta - (n+g+\delta)\,\tilde{h}.}$$

2. (7 points)

Solution:

(a) Setting $\dot{\tilde{k}} = 0$ and $\dot{\tilde{h}} = 0$:

$$s_K\,\tilde{k}^{*\alpha}\tilde{h}^{*\beta} = (n+g+\delta)\,\tilde{k}^*,$$
$$s_H\,\tilde{k}^{*\alpha}\tilde{h}^{*\beta} = (n+g+\delta)\,\tilde{h}^*.$$

Dividing the first by the second:

$$\frac{s_K}{s_H} = \frac{\tilde{k}^*}{\tilde{h}^*} \quad \Longrightarrow \quad \boxed{\tilde{h}^* = \frac{s_H}{s_K}\,\tilde{k}^*.}$$

(b) Substituting $\tilde{h}^* = (s_H/s_K)\tilde{k}^*$ into the first steady-state equation:

$$s_K\,\tilde{k}^{*\alpha}\left(\frac{s_H}{s_K}\right)^\beta \tilde{k}^{*\beta} = (n+g+\delta)\,\tilde{k}^*.$$

Collect powers of s_K and s_H and of $\tilde{k}^*$:

$$s_K \cdot s_K^{-\beta} \cdot s_H^{\beta} \cdot \tilde{k}^{*(\alpha+\beta)} = (n+g+\delta)\,\tilde{k}^*.$$

$$s_K^{1-\beta}\, s_H^{\beta}\, \tilde{k}^{*(\alpha+\beta-1)} = (n+g+\delta).$$

Raise both sides to the power $1/(\alpha+\beta-1)$. Since $\alpha+\beta<1$, the exponent $\alpha+\beta-1<0$, so $1/(\alpha+\beta-1) = -1/(1-\alpha-\beta)$:

$$\tilde{k}^* = \left[\frac{s_K^{1-\beta}\, s_H^{\beta}}{n+g+\delta}\right]^{1/(1-\alpha-\beta)}.$$

Using $\tilde{h}^* = (s_H/s_K)\tilde{k}^*$:

$$\tilde{h}^* = \frac{s_H}{s_K}\left[\frac{s_K^{1-\beta}\, s_H^{\beta}}{n+g+\delta}\right]^{1/(1-\alpha-\beta)} = \left[\frac{s_K^{\alpha}\, s_H^{1-\alpha}}{n+g+\delta}\right]^{1/(1-\alpha-\beta)}.$$

$$\boxed{\tilde{k}^* = \left[\frac{s_K^{1-\beta}\, s_H^{\beta}}{n+g+\delta}\right]^{1/(1-\alpha-\beta)}, \qquad \tilde{h}^* = \left[\frac{s_K^{\alpha}\, s_H^{1-\alpha}}{n+g+\delta}\right]^{1/(1-\alpha-\beta)}.}$$

Steady-state output per effective worker:

$$\tilde{y}^* = \tilde{k}^{*\alpha}\tilde{h}^{*\beta} = \left[\frac{s_K^{1-\beta}\, s_H^{\beta}}{n+g+\delta}\right]^{\alpha/(1-\alpha-\beta)} \cdot \left[\frac{s_K^{\alpha}\, s_H^{1-\alpha}}{n+g+\delta}\right]^{\beta/(1-\alpha-\beta)}.$$

Combining exponents (s_K exponent: $(1-\beta)\alpha/D + \alpha\beta/D = \alpha/D$; s_H exponent: $\beta\alpha/D + (1-\alpha)\beta/D = \beta/D$, where $D = 1-\alpha-\beta$):

$$\boxed{\tilde{y}^* = \left[\frac{s_K^{\alpha}\, s_H^{\beta}}{(n+g+\delta)^{\alpha+\beta}}\right]^{1/(1-\alpha-\beta)}.}$$

3. (7 points)

Solution:

Since $y^* = A_t\,\tilde{y}^*$:

$$\ln y^* = \ln A_0 + g\,t + \frac{\alpha}{1-\alpha-\beta}\ln s_K + \frac{\beta}{1-\alpha-\beta}\ln s_H - \frac{\alpha+\beta}{1-\alpha-\beta}\ln(n+g+\delta).$$

(a) Elasticity of y^* w.r.t. s_K:

- **Basic Solow** ($\beta = 0$): $\varepsilon = \alpha/(1-\alpha)$. For $\alpha = 1/3$: $\varepsilon = 1/2$.
- **Augmented Solow** ($\alpha = \beta = 1/3$): $\varepsilon = (1/3)/(1-1/3-1/3) = (1/3)/(1/3) = 1$.

The elasticity **doubles** from 0.5 to 1.0.

(b) The augmented model amplifies the effect of s_K differences (because $\alpha+\beta > \alpha$ reduces the denominator $1-\alpha-\beta$) and adds a *second channel* through s_H. Cross-country variation in both physical and human capital investment rates can now explain a

much larger fraction of income differences. MRW showed this model can explain roughly 80% of cross-country income variation (vs. ∼60% for basic Solow), substantially reducing the unexplained residual attributed to TFP.

4. (6 points)

Solution:

Key limitation: The model treats s_H (human capital investment rate) as exogenous. In reality, s_H is itself an endogenous choice driven by institutions, returns to education, credit constraints, and government policy. Countries with low s_H may have low s_H *because* of their low income or weak institutions, not as an independent causal factor. This creates an **endogeneity problem**: the model can "explain" income differences with s_H, but s_H itself needs to be explained. A deeper theory would endogenise the education decision (e.g. the Uzawa-Lucas model) to understand *why* s_H differs across countries.
Additionally, TFP (A) is treated as equal across countries. Residual TFP differences (even after adding H) remain large and may reflect institutional quality, misallocation, or technology adoption barriers — factors the augmented Solow model does not address.

2 Question 2 — Ramsey Model: Permanent Increase in TFP Growth

1. (5 points)

Solution:

(a) Law of motion:

$$\dot{\tilde{k}} = \tilde{k}^{\alpha} - \tilde{c} - (n + x + \delta)\,\tilde{k}.$$

(b) Euler equation (per-effective-worker):

$$\frac{\dot{\tilde{c}}}{\tilde{c}} = \frac{1}{\sigma}\big(\alpha\tilde{k}^{\alpha-1} - \delta - \rho - \sigma x\big).$$

(c) Loci:

- $\dot{\tilde{c}} = 0$: vertical line at $\tilde{k}^*$ where $\alpha\tilde{k}^{*(\alpha-1)} = \delta + \rho + \sigma x$. To the left, $\dot{\tilde{c}} > 0$; to the right, $\dot{\tilde{c}} < 0$.
- $\dot{\tilde{k}} = 0$: $\tilde{c} = \tilde{k}^{\alpha} - (n + x + \delta)\tilde{k}$ (hump-shaped curve). Above it, $\dot{\tilde{k}} < 0$; below, $\dot{\tilde{k}} > 0$.

The steady state is at the intersection. The saddle path approaches it from both sides.

2. (7 points)

Solution:

(a) Modified Golden Rule: $\alpha\tilde{k}^{*(\alpha-1)} = \delta + \rho + \sigma x$.

$$\boxed{\tilde{k}^* = \left[\frac{\alpha}{\delta + \rho + \sigma x}\right]^{1/(1-\alpha)}}.$$

$$\tilde{c}^* = \tilde{k}^{*\alpha} - (n + x + \delta)\,\tilde{k}^*.$$

(b) Modified Golden Rule: $f'(\tilde{k}^*) = \delta + \rho + \sigma x$.
(c) The TVC (in per-effective-worker form) is $\lim_{t\to\infty} \lambda(t)\,\tilde{k}(t)\,e^{-\rho' t} = 0$, where $\rho' = \rho - n - (1-\sigma)x$.
On the BGP, $\tilde{k}$ is constant (growth rate 0), and from the FOC $\lambda = \tilde{c}^{-\sigma}$ with $\tilde{c}$ growing at rate $\dot{\tilde{c}}/\tilde{c} = (1/\sigma)(\alpha\tilde{k}^{*(\alpha-1)} - \delta - \rho - \sigma x)$. On the BGP this equals 0 (since we are at the steady state $\tilde{k}^*$), so λ is constant. Thus $\lambda(t)\tilde{k}(t) \to$ const, and the TVC requires $e^{-\rho' t} \to 0$, i.e. $\rho' > 0$:

$$\rho' = \rho - n - (1-\sigma)x > 0 \quad \Longleftrightarrow \quad \rho - n > (1-\sigma)x.$$

This ensures that the discount factor dominates growth in the costate and that lifetime utility is bounded (otherwise the household's objective integral diverges).

3. (7 points)

Solution:

(a) New steady state with $x' > x$:

$$\tilde{k}^{*\prime} = \left[\frac{\alpha}{\delta + \rho + \sigma x'}\right]^{1/(1-\alpha)} < \tilde{k}^* = \left[\frac{\alpha}{\delta + \rho + \sigma x}\right]^{1/(1-\alpha)}$$

since $\delta+\rho+\sigma x' > \delta+\rho+\sigma x$ and the exponent $1/(1-\alpha) > 0$, so the expression decreases. For per-effective-worker consumption, use the $\dot{\tilde{k}} = 0$ locus evaluated at the steady state:

$$\tilde{c}^{*\prime} = \tilde{k}^{*\prime\alpha} - (n + x' + \delta)\,\tilde{k}^{*\prime} = \tilde{k}^{*\prime}\big[\tilde{k}^{*\prime\alpha-1} - (n + x' + \delta)\big].$$

Comparing with $\tilde{c}^* = \tilde{k}^*\big[\tilde{k}^{*\alpha-1} - (n + x + \delta)\big]$, there are two reinforcing effects both reducing $\tilde{c}^{*\prime}$:

(i) **Lower** $\tilde{k}^{*\prime}$: since $\tilde{k}^{*\prime} < \tilde{k}^*$, the production term $\tilde{k}^{*\prime\alpha} < \tilde{k}^{*\alpha}$.

(ii) **Higher effective depreciation**: $(n + x' + \delta) > (n + x + \delta)$, so more output is required to keep $\tilde{k}$ constant.

Both channels reduce $\tilde{c}^{*\prime}$ below $\tilde{c}^*$. Indeed, since $\tilde{k}^{*\prime}$ satisfies the Modified Golden Rule $\alpha\tilde{k}^{*\prime\alpha-1} = \delta + \rho + \sigma x'$, we can also write $\tilde{c}^{*\prime} = \tilde{k}^{*\prime\alpha}(1 - (n + x' + \delta)/\tilde{k}^{*\prime\alpha-1} \cdot \alpha^{-1} \cdot \alpha)$; the key point is that $\tilde{k}^{*\prime}$ is strictly smaller, and the $\dot{\tilde{k}} = 0$ locus is shifted down, confirming $\tilde{c}^{*\prime} < \tilde{c}^*$.

Intuition: Faster TFP growth means that per-effective-worker capital depreciates faster (the "effective worker" grows faster), so the economy sustains a lower level of $\tilde{k}$ in steady state. The household also requires a higher return to save (to compensate for faster-growing consumption), which further discourages capital accumulation in effective units.

(b) Shifts:

- $\dot{\tilde{c}} = 0$ shifts **left**: from $\tilde{k}^*$ to $\tilde{k}^{*\prime} < \tilde{k}^*$.
- $\dot{\tilde{k}} = 0$ shifts **down**: higher $(n + x' + \delta)$ steepens the effective depreciation line, lowering the curve $\tilde{c} = \tilde{k}^\alpha - (n + x' + \delta)\tilde{k}$.

(c)

- $\tilde{k}$ does **not jump**: K, A, and L are all continuous at $t = 0$ (only $\dot{A}/A$ changes, not A itself).
- $\tilde{c}$ **jumps up**: the household immediately raises per-effective-worker consumption to reach the new saddle path. Since $\tilde{k}(0) = \tilde{k}^* > \tilde{k}^{*\prime}$, the economy is to the right of the new steady state.
- **Transition**: both $\tilde{c}$ and $\tilde{k}$ fall gradually along the new saddle path from $(\tilde{k}^*, \tilde{c}(0^+))$ to $(\tilde{k}^{*\prime}, \tilde{c}^{*\prime})$.

4. (6 points)

Solution:

Per-capita consumption and income:

$$c_t = \tilde{c}_t\, A_t, \qquad y_t = \tilde{y}_t\, A_t.$$

Before the shock, c_t and y_t grow at rate x. After the shock, they grow at rate $x' > x$ in the long run (since $\tilde{c}_t \to \tilde{c}^{*\prime}$ and A_t grows at x').
Although $\tilde{c}^{*\prime} < \tilde{c}^*$ (the per-effective-worker *level* falls), the per-capita *growth rate* is permanently higher. At some finite time, the new path overtakes the old one:

$$\tilde{c}^{*\prime}\, A_0 e^{x't} > \tilde{c}^*\, A_0 e^{xt} \quad \text{for all } t > \frac{\ln(\tilde{c}^*/\tilde{c}^{*\prime})}{x' - x}.$$

The household is better off because the present discounted value of lifetime utility is higher: faster permanent growth outweighs the short-run reduction in consumption level. Faster technological progress is unambiguously welfare-improving.

3 Question 3 — Development Accounting and Misallocation

1. (5 points)

Solution:

(a) Take logs: $\ln Y = \ln A + \alpha \ln K + (1-\alpha)\ln L$.
Differentiate w.r.t. time:

$$\boxed{\gamma_Y = \gamma_A + \alpha\,\gamma_K + (1-\alpha)\,\gamma_L.}$$

The **Solow residual** is: $\gamma_A = \gamma_Y - \alpha\gamma_K - (1-\alpha)\gamma_L$. It captures everything that contributes to output growth beyond measured factor accumulation — including technology, efficiency, measurement error, and institutional change.
(b) Two factors besides technology: (i) **misallocation** of resources across firms/sectors (changes in allocative efficiency show up as TFP changes); (ii) **mismeasurement** of input quality — e.g. not adjusting labour for human capital or capital for quality improvements.

2. (7 points)

Solution:

In per-worker terms: $y = Ak^\alpha$.
Taking ratios: $\ln(y_R/y_P) = \ln(A_R/A_P) + \alpha\ln(k_R/k_P)$.
With $y_R/y_P = 27$, $k_R/k_P = 27$, $\alpha = 1/3$:
(a) Capital contribution: $\alpha\ln(k_R/k_P) = \frac{1}{3}\ln 27 = \frac{1}{3}\cdot 3\ln 3 = \ln 3$.
As a fraction of $\ln(y_R/y_P) = \ln 27 = 3\ln 3$:

$$\text{Capital share} = \frac{\ln 3}{3\ln 3} = \boxed{\frac{1}{3} \approx 33\%.}$$

(b) TFP contribution (residual): $\ln(A_R/A_P) = 3\ln 3 - \ln 3 = 2\ln 3$.

$$\text{TFP share} = \frac{2\ln 3}{3\ln 3} = \boxed{\frac{2}{3} \approx 67\%.}$$

(c) TFP differences are the **dominant** proximate source of cross-country income differences, accounting for about two-thirds of the gap. This motivates studying what drives TFP: technology adoption, institutions, and misallocation.

3. (7 points)

Solution:

(a) Efficient allocation equalises MPK across firms: $\alpha A k_1^{\alpha-1} = \alpha A k_2^{\alpha-1} \Rightarrow k_1 = k_2 = K/2$.

$$Y^{\text{eff}} = 2A(K/2)^\alpha = 2^{1-\alpha} A\,K^\alpha.$$

(b) With the subsidy, firm 1 optimises taking its effective rental cost as $(1-\tau)r$; firm 2

takes r. Profit-maximising capital demands equate MPK to effective rental cost:

$$\begin{aligned} \text{Firm 1:} \quad & \alpha A k_1^{\alpha-1} = (1-\tau)r, \\ \text{Firm 2:} \quad & \alpha A k_2^{\alpha-1} = r. \end{aligned}$$

Divide the first equation by the second (αA cancels):

$$\frac{k_1^{\alpha-1}}{k_2^{\alpha-1}} = (1-\tau) \quad \Longrightarrow \quad \left(\frac{k_1}{k_2}\right)^{\alpha-1} = (1-\tau).$$

Raise both sides to the power $\frac{1}{\alpha-1}$. Since $\alpha < 1$, $\alpha - 1 < 0$, so $\frac{1}{\alpha-1} = -\frac{1}{1-\alpha}$:

$$\frac{k_1}{k_2} = (1-\tau)^{1/(\alpha-1)} = (1-\tau)^{-1/(1-\alpha)}.$$

Because $\tau \in (0,1)$, we have $(1-\tau) \in (0,1)$, so $(1-\tau)^{-1/(1-\alpha)} > 1$.

$$\boxed{\frac{k_1}{k_2} = (1-\tau)^{-1/(1-\alpha)} > 1.}$$

Firm 1 (subsidised) receives *too much* capital relative to the efficient allocation.
(c) Aggregate output with misallocation: $Y^{\text{mis}} = A(k_1^\alpha + k_2^\alpha)$ with $k_1 \neq k_2$ but $k_1 + k_2 = K$.
Since $f(k) = k^\alpha$ is **strictly concave** (for $\alpha < 1$), Jensen's inequality gives:

$$\frac{k_1^\alpha + k_2^\alpha}{2} < \left(\frac{k_1 + k_2}{2}\right)^\alpha = (K/2)^\alpha$$

whenever $k_1 \neq k_2$. Therefore:

$$Y^{\text{mis}} = A(k_1^\alpha + k_2^\alpha) < 2A(K/2)^\alpha = Y^{\text{eff}}. \quad \checkmark$$

Intuition: The subsidy channels too much capital to firm 1, where the marginal product is low, and too little to firm 2, where the marginal product is high. Aggregate output would be higher if capital were reallocated from firm 1 to firm 2 until marginal products are equalised. This is the **essence of misallocation**: dispersion in marginal products signals that a reallocation of inputs could increase aggregate output without using more resources.

4. (6 points)

Solution:

(a) Misallocation is defined as the dispersion of marginal revenue products across firms: $\text{Var}(\text{MRPK}_i) > 0$. In an efficient economy with no frictions, all firms equalise MRPK to the common rental rate r, so $\text{Var}(\text{MRPK}) = 0$. Positive variance means that some firms have higher returns to capital than others, implying that total output could be increased by reallocating capital from low-MRPK firms to high-MRPK firms.
(b) Two concrete examples of policies/frictions creating capital wedges:

(i) **Directed/subsidised credit:** State-owned banks in China provide below-market loans to state-owned enterprises (SOEs), giving them artificially low capital costs.

Private firms face higher borrowing rates ⇒ MRPK dispersion.

(ii) **Size-dependent regulations:** In India, labour laws (e.g. the Industrial Disputes Act) impose high firing costs on firms above a size threshold. This discourages productive firms from growing to efficient scale, keeping capital trapped in smaller, less productive establishments.

Other valid examples: trade protection for specific sectors, entry barriers from licensing, corruption/connections determining access to resources, land-use regulations preventing reallocation.

4 Short Questions

1. (9 points) — Golden Rule vs. Modified Golden Rule

Solution:

(a) In the Solow model, steady-state consumption is $c^* = f(\tilde{k}^*) - (n+g+\delta)\tilde{k}^*$. Maximising over $\tilde{k}^*$:

$$\frac{\mathrm{d}c^*}{\mathrm{d}\tilde{k}^*} = f'(\tilde{k}^*_{GR}) - (n+g+\delta) = 0 \quad \Longrightarrow \quad \boxed{f'(\tilde{k}^*_{GR}) = n+g+\delta.}$$

This is *independent of* s because s determines which steady state the economy reaches, but the Golden Rule picks the $\tilde{k}^*$ that maximises c^* regardless of how you get there.
(b) In the Ramsey model, the Modified Golden Rule is: $f'(\tilde{k}^*) = \delta + \rho + \sigma x$.
This generally differs from the Golden Rule because the Ramsey household is *impatient* ($\rho > 0$): it discounts future consumption, so it does not save enough to reach the Golden Rule level. The two coincide if and only if $\rho + \sigma x = n + g$, i.e. $\rho = n + (1-\sigma)g$ (with $g = x$), which is a knife-edge condition.
(c) The Solow economy *can* be dynamically inefficient: if s is high enough that $\tilde{k}^* > \tilde{k}^*_{GR}$, the economy over-accumulates capital (MPK $< n + g + \delta$). Reducing s would raise consumption in every period.
The Ramsey economy *cannot* be dynamically inefficient: optimising households never accumulate beyond the Modified Golden Rule, which satisfies $f'(\tilde{k}^*) > n + g + \delta$ (given the TVC constraint $\rho - n > (1-\sigma)x$). Dynamic efficiency is guaranteed by forward-looking optimisation.

2. (8 points) — Role of Monopolistic Competition

Solution:

(a) Ideas are **non-rival**: once created, they can be used by everyone at zero marginal cost. Under perfect competition, price equals marginal cost, so $p = MC = 0$ for ideas. But then innovators earn zero revenue and cannot recoup the fixed cost of R&D. No firm would invest in innovation $\Rightarrow$ no endogenous growth. The fundamental problem: non-rivalry requires increasing returns, which is incompatible with perfect competition.
(b) Monopolistic competition gives innovators *temporary market power*. In the Romer model, the markup $p = 1/\alpha > 1$ (above marginal cost of production) generates positive profits $\pi > 0$. These profits are the *reward for innovation* and are essential for financing R&D. Without the markup, the value of creating a new variety $V = \pi/r = 0$, and no one would innovate.
(c) The markup creates a **static inefficiency**: monopolists restrict output below the socially optimal level ($x^{\text{mkt}} < x^{\text{plnr}}$). A **subsidy on intermediate purchases** equal to $1-\alpha$ (so the effective price faced by final good producers falls from $1/\alpha$ to 1) corrects this distortion. This raises intermediate usage to the efficient level while preserving the innovator's profit incentive (profits are replaced by the subsidy).

3. (8 points) — Expanding Varieties vs. Quality Ladders

Solution:

(a) In Romer, new varieties *coexist* with old ones — no firm is displaced. In Aghion–Howitt, new quality improvements *replace* old producers. Key implication: in quality ladders, incumbent monopolists **lose their rents** when a successor innovates. This creates a direct conflict between incumbents and entrants that is absent in the expanding varieties framework.

(b) **Creative destruction**: the process by which new innovators displace incumbents. The **business-stealing externality**: the entrant captures the entire market, destroying the incumbent's profit stream, but does not internalise this cost. This leads to **over-innovation** relative to the social optimum, because the private return to innovation includes "stealing" rents that were already generating social value.

(c) The **appropriability externality**: the innovator cannot capture all the social value of the innovation — consumers benefit from quality improvements beyond what monopoly profits reflect (consumer surplus). This pushes toward **under-innovation**.

The two externalities act in *opposite* directions. The net effect is ambiguous and depends on parameters. This means the market innovation rate can be either too high or too low relative to the social optimum. Optimal policy requires balancing both externalities, potentially with multiple instruments (R&D subsidies/taxes + output subsidies).

End of Answer Key — Practice Exam 2.